

HW 5

1) a) From the zoo, we recognize this could be a uniform distribution, because the dice is fair and rolls integers.

$$E(X) = \frac{a+b}{2} = \frac{1+10}{2} = \boxed{11/2}$$

$$\text{Var}(X) = \frac{(b-a)(b-a+1)}{12} = \frac{(9)(10)}{12} = \frac{90}{12} = \boxed{\frac{33}{4}}$$

$$b) E(Y) = \sum_{i=1}^8 E(X_i)$$

$$= 8 \cdot \frac{11}{2} = \boxed{44}$$

Because each dice roll is independent,

$$\text{Var}(Y) = \sum_{i=1}^8 \text{Var}(X_i)$$

$$= 8 \cdot \left(\frac{33}{4}\right) = 2 \cdot 33 = \boxed{66}$$

we can sum them

together like $\sum_{i=1}^8 E(X_i)$ or $\sum_{i=1}^8 \text{Var}(X_i)$.